

Algebra 1
Unit 8
Lesson 2 Practice

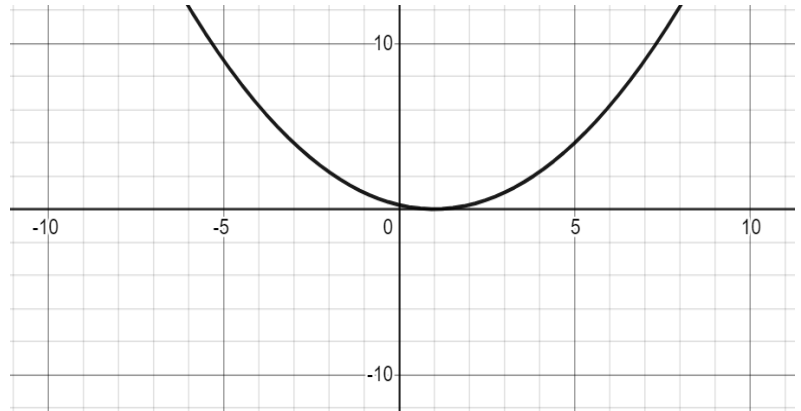
Name **Answer Key** _____
Date _____
Period _____

1. As the x -values of the parabola get infinitely *larger*, the y -values of the parabola approach

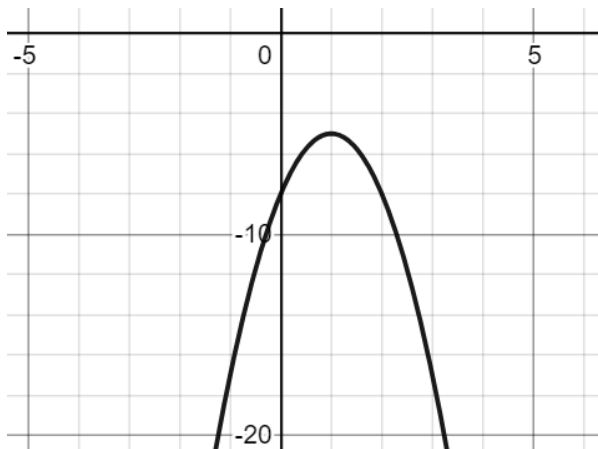
☒ ∞
☐ $-\infty$

2. As the x -values of the parabola get infinitely *smaller*, the y -values of the parabola approach

☒ ∞
☐ $-\infty$



Use the graph of the quadratic for questions 3 and 4.



3. Use words or symbols to describe the domain of the quadratic.

$$-\infty < x < \infty$$

4. Use words or symbols to describe the range of the quadratic.

$$-\infty < y \leq -5$$

5. Determine the x -intercepts and y -intercepts of a quadratic using the equation

$$y = -\frac{4}{3}(x - 2)(x + 3).$$

$$0 = -\frac{4}{3}(x - 2)(x + 3)$$

$$0 = (x - 2)(x + 3)$$

$$x = 2$$

$$x = -3$$

$$(2, 0)(-3, 0)$$

$$y = -\frac{4}{3}(0 - 2)(0 + 3)$$

$$y = -\frac{4}{3}(-2)(3)$$

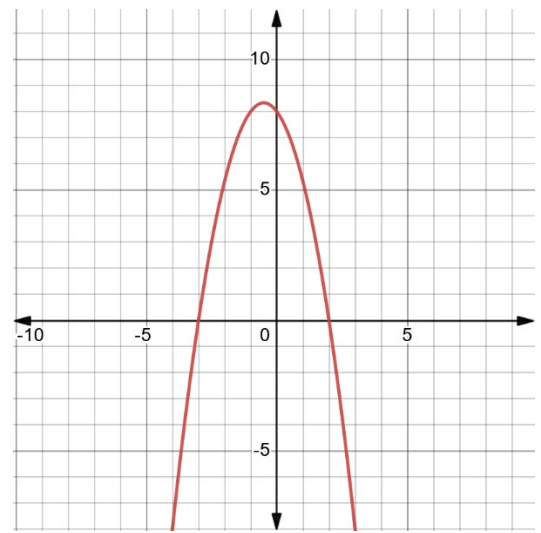
$$y = 8 \quad (0, 8)$$

6. Use the table of values to determine the vertex of the parabola.

$$\text{Vertex} = (-0.5, 8.333)$$

x	y
-2	5.333
-1.5	7
-1	8
-0.5	8.333
0	8
0.5	7
1	5.333

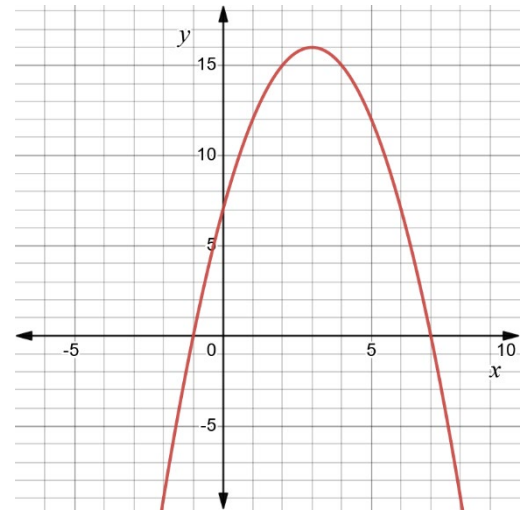
7. Use the key features you determined in question 5 and 6 to graph the parabola of the quadratic function.



8. Use the equation $y = -(x - 7)(x + 1)$ to graph the quadratic.

$$\begin{array}{ll}
 0 = -(x - 7)(x + 1) & y = -(0 - 7)(0 + 1) \\
 0 = (x - 7)(x + 1) & 0 = -(-7)(1) \\
 x = 7 \quad x = -1 & y = 7 \\
 (7, 0) \quad (-1, 0) & (0, 7)
 \end{array}
 \quad
 \begin{array}{c|c}
 x & y \\
 \hline
 1 & 12 \\
 2 & 15 \\
 3 & 16 \\
 4 & 15
 \end{array}$$

Vertex = (3, 16)



9. Determine all of the key features possible using the equation of the quadratic $y = 2(x - 2)(x - 1)$.

$$\begin{array}{ll}
 0 = 2(x - 2)(x - 1) & y = 2(0 - 2)(0 - 1) \\
 0 = (x - 2)(x - 1) & y = 2(-2)(-1) \\
 x = 2 \quad x = 1 & y = 4 \\
 (2, 0) \quad (1, 0) & (0, 4)
 \end{array}
 \quad
 \begin{array}{c|c}
 x & y \\
 \hline
 0 & 4 \\
 0.5 & 1.5 \\
 1 & 0 \\
 1.5 & -0.5 \\
 2 & 0
 \end{array}$$

x -intercepts y -intercept

vertex, axis of symmetry

10. Use the key features found in question 9 to graph the quadratic.

